



Experimental and modeling study of hex-5-en-1-yl radical pyrolysis at very high pressure and temperature

Mirosław K. Liszka, Kenneth Brezinsky*

Department of Mechanical and Industrial Engineering, University of Illinois at Chicago, Chicago, IL 60607, USA

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ABSTRACT

The pyrolysis of the hex-5-en-1-yl radical was studied with the use of 6-bromo-1-hexene as the precursor. Shock tube experiments were completed at high pressures (40, 100, and 200 bar) and temperatures (900 to 1450 K) with dilute mixtures of 6-bromo-1-hexene (approximately 160 ppm) to determine whether a variation of the experimental pressure has an effect on the product distribution, and whether high experimental pressures may lead to the formation of alkylcyclopentanes. Another set of experiments was completed using a higher concentration fuel mixture of 6-bromo-1-hexene (358 ppm) to determine whether the product distribution changes were caused by varying the experimental pressure, or the initial fuel concentration in the reaction zone. It was found that at the current experimental conditions the product distribution is sensitive to the initial fuel concentration. Among the major products observed in the current work were ethylene, propadiene, propyne, methane, acetylene, propene, diacetylene, vinylacetylene, 1,5-hexadiene, benzene, and cyclohexene. An additional set of experiments was also completed using 1,5-hexadiene as the fuel to determine whether 6-bromo-1-hexene pyrolysis might be dominated by HBr loss which would result in the formation of 1,5-hexadiene as the dominant initial product, but it was found not to be the case. A mechanism was generated using the Reaction Mechanism Generator with the JetSurF 2 mechanism as the seed mechanism in an attempt to model the current experimental data, but it was found to not be able to capture the benzene formation in the 6-bromo-1-hexene experiments. A reasonable prediction of the 6-bromo-1-hexene pyrolysis data was obtained with the mechanism if the fuel mixture used for simulations was composed of 50% 1,5-hexadiene and 50% hex-5-en-1-yl radical. Using purely hex-5-en-1-yl radical as the fuel for the simulations resulted in poor prediction of the experimental data for almost all the species, except for methane.

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1. Introduction

The hex-5-en-1-yl radical plays an important role in the dissociation of alkylcyclohexanes. An example illustrating this has been presented in a laminar flame speed study by Ji et al., which used cyclohexane and alkylcyclohexanes up to n-butylcyclohexane as fuels in the comparison of three published mechanisms against the experimental data [1]. The mechanisms used were JetSurF1.1 [2], Dagaut and Cathonnet's kerosene model [3], and Silke et al., cyclohexane oxidation model [4]. In flames, all three mechanisms predicted that approximately 95% of cyclohexane is converted into cyclohexyl radicals by H abstractions. Dagaut's and Silke's model both predicted that approximately 95% of cyclohexane then isomerizes to the hex-5-en-1-yl radical, and JetSurF1.1 predicted that 73% of cyclohexyl radical will isomerize to hex-5-en-1-yl. JetSurF1.1 was also validated against the shock tube ignition data of Hong et

al., [5], with similar results obtained in the cyclohexane and cyclohexyl dissociation analysis. A comparative study comparing multiple literature mechanism against newly obtained experimental data, much like the work by Ji et al., [1], was completed by Liu et al., which investigated the flame ignition temperatures of cyclohexane and alkylcyclohexanes up to n-butylcyclohexane [6], and compared the experimental results against the JetSurF2.0 mechanism [7] and Dagaut's mechanism [3] with the finding being in agreement in regards to the branching of cyclohexane and cyclohexyl dissociation. Two other modeling efforts of cyclohexane flames by McEnally and Pfefferle [8] and Zhang et al., [9] have likewise proposed the reaction pathway of cyclohexane dissociation to cyclohexyl radical which then isomerizes to hex-5-en-1-yl as the major reaction pathway. Other studies also emphasize the importance of the hex-5-en-1-yl radical [1,10–12] and its ability to form the cyclopentylmethyl [12–14] radical which is expected to be the precursor for methylcyclopentane, an alkylcyclopentane.

Alkylcyclopentanes have been found to be the dominant products in the pyrolysis of alkylcyclohexanes in the supercritical

* Corresponding author.

E-mail address: kenbrez@uic.edu (K. Brezinsky).

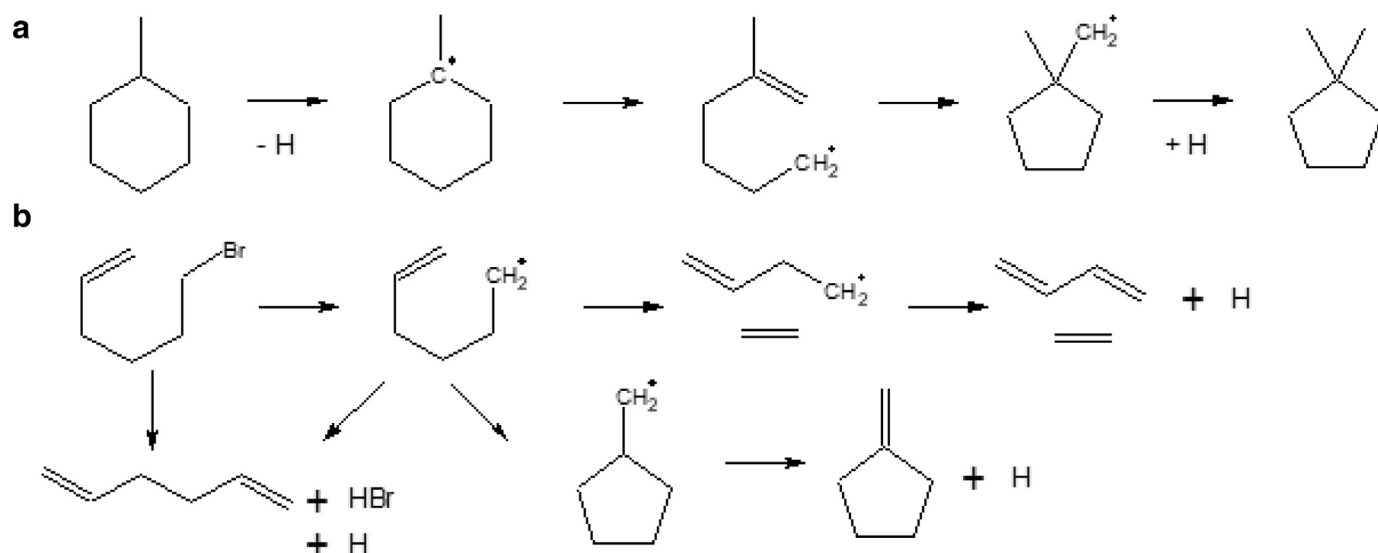


Fig. 1. (a) Methylcyclohexane reaction pathway leading to 1,1-dimethylcyclopentane formation [16], (b) three competing 6-bromo-1-hexene decomposition reaction pathways leading to the formation of 1,5-hexadiene, 1,3-butadiene and ethylene, and methylenecyclopentane.

phase (pressures exceeding the critical pressure and up to 100 bar, at temperatures of around or below 800 K) [15–17] with the proposed reaction paths analogous to the formation of methylcyclopentane in cyclohexane pyrolysis [15], i.e. cyclohexane $\rightarrow$ cyclohexyl radical $\rightarrow$ hex-5-en-1-yl radical $\rightarrow$ cyclopentylmethyl radical $\rightarrow$ methylcyclopentane [16]. The reaction pathway involving the hex-5-en-1-yl radical is additionally important in the pyrolysis of alkylcyclohexanes because the same reaction pathway is possible after the cleavage of the alkyl group from the alkylcyclohexane fuel resulting in the formation of cyclohexyl radical [5,18]. Alkylcyclopentanes have also been found to be both sooting [19] and coking [17] precursors; therefore, gaining a better understanding of the conditions needed for their formation is beneficial to the optimization of the performance of combustion systems.

Despite the importance of hex-5-en-1-yl in the chemical kinetic modeling of combustion and pyrolysis of cyclohexane, and alkylcyclohexanes, there has only been one experimental study conducted at combustion relevant temperatures seeking to investigate the pyrolysis of hex-5-en-1-yl, which is the work by Peukert et al., [12] utilizing 6-iodo-1-hexene as the precursor to hex-5-en-1-yl. Additionally, the work by Peukert et al., encompassed a 100 K temperature range from 1060 to 1160 K, at a pressure of about 2 bar. In the study the H atom profiles were measured and iodine-ARAS was used to estimate the rate constant for the initial bond fission 6-iodo-1-hexene to I + hex-5-en-1-yl radical. It is the aim of this study to investigate the pyrolysis of hex-5-en-1-yl by using 6-bromo-1-hexene as the precursor to quantify the stable species products at target, or nominal, pressures of 40, 100, and 200 bar encompassing a temperature range of approximately 900 to 1450 K. Figure 1a) provides one of the proposed pathways for the formation of alkylcyclopentanes in the supercritical phase experiments [16]. Figure 1b) illustrates an analogous pathway that begins with 6-bromo-1-hexene. The pathway in Fig. 1b) is also a pathway that can occur in the pyrolysis of cyclohexane. Cyclohexane would first undergo hydrogen loss forming the cyclohexyl radical that upon ring opening would form the hex-5-en-1-yl radical. The benefit of using 6-bromo-1-hexene to investigate this pathway is that the hex-5-en-1-yl radical can be obtained immediately following the bromine loss from 6-bromo-1-hexene so the competing reaction pathways of cyclohexane dissociation (such as isomerization to 1-hexene) or the cyclohexyl

radical dissociation (such as the formation of cyclohexene) will not occur. The drawback is that 6-bromo-1-hexene may also undergo HBr loss resulting in the formation of 1,5-hexadiene. For the pathway in Fig. 1b) to be truly analogous with that shown in Fig. 1a) the final product of hex-5-en-1-yl radical ring closure should be methylcyclopentane; however, the final product shown is methylenecyclopentane. The prediction of methylenecyclopentane being the final product of hex-5-en-1-yl radical ring closure is based on our recent high pressure cyclohexane and methylcyclohexane study in which only methylenecyclopentane was quantified, in trace amounts, and no alkylcyclopentanes were observed to form [20,21]. Due to the high pressures present in the present work it will be possible to make a direct comparison of the product distribution with those observed in the supercritical phase experiments [15–17] on the basis of the experimental pressure, and whether an abundance of alkylcyclopentanes will be observed to form in the gas phase.

2. Experimental apparatus

The design and operating procedure of the High Pressure Shock Tube (HPST) has been published and described in detail [22] with only a brief description to follow. The HPST is a single pulse shock tube composed of a 120-inch-long driven section and a 60-inch-long driver section. The HPST is a convergent shock tube with a driver and driven bore of 2 and 1 inches, respectively. A 3003 aluminum diaphragm is used to separate the high pressure driver gas from the low pressure driven gas. The diaphragm is scored to facilitate clean opening of the diaphragm. The presented work was completed at three different pressures requiring three different thicknesses of diaphragms as well as different depths of the score. The nominal thicknesses of the diaphragms used to obtain nominal reflected shock pressures of 40, 100, and 200 bar in the end wall reaction zone were: 25 thousandths of an inch diaphragm thickness with 14 thousandths of an inch wall thickness, the wall thickness is the thickness of the material in the score, 32 thousandths of an inch diaphragm thickness with 27 thousandths of an inch wall thickness, and 50 thousandths of an inch diaphragm thickness with 43 thousandths of an inch wall thickness, respectively. The performance of the diaphragms was found to vary slightly between the batches of the aluminum stock from

which the diaphragms were made. To achieve more consistent reflected shock pressures the wall thickness would be adjusted by half or one thousandth of an inch from the nominal wall thickness. All the diaphragms were made on campus at the University of Illinois at Chicago engineering machine shop.

The temperature in the reaction zone was not measured directly but by the means of a chemical thermometer [23]. The extent of reaction of the chemical thermometer is used to back out the temperature which is then calibrated against the incident shock speed. The incident shock speed is measured with six side-wall mounted PCB-113A series transducers near the end wall of the shock tube. A seventh transducer, also PCB-113A series, is mounted in the end wall of the shock tube and is used to obtain the reflected shock pressure trace at the end wall. The end wall pressure trace is used to obtain both the pressure in the reaction zone, and the reaction time by measuring the time from the arrival of the shock wave to the time at which the pressure in the reaction zone drops to 80% [23,24]. The chemical thermometers used in the present study were cyclopropanecarbonitrile (Sigma-Aldrich, 98%) and 1,1,1-trifluoroethane (Airgas, 99%). Cyclopropanecarbonitrile was used to calibrate the temperature over a range of 1000 to 1100 K, and 1,1,1-trifluoroethane was used to obtain the temperature calibration over approximately 1300 to 1400 K. The extents of decomposition for both the thermometers were between 10 and 90 percent in order to obtain a linear relationships between the extents of decomposition and temperature. A calibration curve was generated using both the chemical thermometers so that the incident shock wave speeds resulting in temperatures falling in a range between those covered by the two chemical thermometers were interpolated, and those resulting in temperatures below or above both the chemical thermometers were extrapolated. The reaction rate coefficients used for cyclopropanecarbonitrile were those determined by Lifshitz et al., [25], and the 1,1,1-trifluoroethane reaction rate coefficients were taken from the work by Matsugi et al., [26] A 30 K uncertainty in the temperature using the current temperature calibration techniques has been estimated in previous works [27,28].

After the arrival of the rarefaction wave at the end wall and quenching of the reaction zone a sample is extracted and analyzed by the means of online gas chromatography sampling which has been described in detail by Comandini et al., [29] Two columns, Supelco Petrocol (24,160-U) and Agilent GS-GasPro (113–4362), were used to identify the stable product species in the experiments. The species quantified in the present work have been identified using test gas calibration mixtures supplied by Air Liquide and a Hewlett–Packard 5973 Mass Selective Detector. After all the species of interest were identified, flame ionization detectors (FID) were calibrated and used to detect and quantify the species in the experiments. All the species in the present work were calibrated with the test gas calibration mixtures except 1,5-hexadiene, 1,3-cyclopentadiene, 1,3-cyclohexadiene and cyclohexene which were calibrated by vaporizing a known amount of a liquid sample of the respective species. The 1,5-hexadiene (Sigma-Aldrich, 97%), 1,3-cyclopentadiene, 1,3-cyclohexadiene (Sigma-Aldrich, 97%), and cyclohexene (Sigma-Aldrich, 99%) species data is estimated to have uncertainty of up to 10%, while all the remaining species have an uncertainty of 5% as specified by Air Liquide. 6-bromo-1-hexene was supplied by Sigma-Aldrich (95%) and the argon and helium used for the mixture and as driven and driver gases were supplied by Praxair (UHP-99.999%).

The test gas mixtures were prepared in a 1.5 ft³ stainless steel cylinder. The cylinder was heated to 100°C using HTS/Amptek heating tape to avoid condensation of the fuel species. An Edwards RV8 vacuum pump was used to bring the cylinder pressure down to approximately 20 mTorr. The fuel was first transferred to a glass bulb in which it was degassed through a freeze-thaw process while kept

under vacuum. The process is repeated at least three times, after which the fuel is allowed to evaporate into the cylinder and the pressure was monitored. Once an adequate pressure of the fuel inside the cylinder is achieved, the cylinder would be filled to the final pressure with the bath gas, argon. Following the mixture preparation process, the mixtures were allowed to homogenize overnight (at least 12 h). The following day the mixtures are sampled to confirm that the desired concentration of fuel was achieved, and that no contamination of the mixture occurred, and if no problems occurred the mixtures were used for experiments.

3. Effects of pressure on 6-bromo-1-hexene pyrolysis

Three sets of shock tube pyrolysis experiments were conducted with 6-bromo-1-hexene as the fuel at three nominal pressures. The target pressures, with an experimental pressure spread in the actual pressures obtained, were 40, 100, and 200 bar with a mean reaction times of 2.68, 2.44, and 2.24 ms, respectively. The mole fraction of 6-bromo-1-hexene in the test gas mixture was 164, 162, and 161 ppm for the 40, 100, and 200 bar experiments, respectively. The mole fraction of the fuel in the test gas was obtained by GC sampling of the test gas mixture after it was allowed to homogenize at least 12 h post preparation allowing it to be ready for use in the experiments.

Figure 2 shows the species mole fractions as a function of temperature of the fuel, 6-bromo-1-hexene, and the major species present in the experiments: ethylene, propadiene, propyne, methane, acetylene, propene, diacetylene, vinylacetylene, 1,5-hexadiene, benzene, and cyclohexene. All the minor species which were quantified have their profile plots included in Figure S1 in the supplemental material. Those species include: ethane, propane, toluene, cyclopropane, 1-butene, isobutylene, 1,3-cyclopentadiene, 1-hexene, and 1,3-cyclohexadiene. 1,3-Butadiene was also expected to form based on the pathway shown in Fig. 1b), but 1,3-butadiene was not successfully quantified in any of the 6-bromo-1-hexene experiments. There was significant scatter in the 1,3-butadiene data, and no clear formation profiles emerged. This problem with 1,3-butadiene recovery occurred in all the 6-bromo-1-hexene experiments, but not in the subsequently conducted 1,5-hexadiene experiments or the cyclohexane [20] and methylcyclohexane [21] experiments which were published separately. The scatter was expected to be caused by bromine containing species with a similar retention time to 1,3-butadiene. Due to the average carbon recovery in the 6-bromo-1-hexene experiments exceeding 90%, 1,3-butadiene was not expected to be a major species even if it constituted all the unrecovered carbon.

The amount of ethane, ethylene, propane, and toluene recovered decreased with a decrease in the experimental pressure. Both ethane and propane show clearly different peak amounts at different experimental pressures. For ethylene and toluene the 100 and 200 bar profiles are similar, but the 40 bar profile shows lower amounts recovered. For propadiene and propyne the amount recovered decreased with increasing experimental pressure, although the 40 and 100 bar profiles are very close and only the 200 bar profiles show less of the two species being recovered. Methane, acetylene, and propene were the three most abundant products, and their profiles did not seem exhibit any differences due to the variation in experimental pressure. Two other substantial products were 1,5-hexadiene and benzene, with their profiles being unaffected by the variation in experimental pressure. The production of 1,5-hexadiene was expected due to 6-bromo-1-hexene undergoing HBr loss. The large amounts of benzene recovered in the lower experimental temperature range were unexpected, because the hex-5-en-1-yl radical is primarily expected to dissociate into fragmented species. The benzene profile initially peaks at approximately 1150 K at then plateaus with an additional increase in the amount of

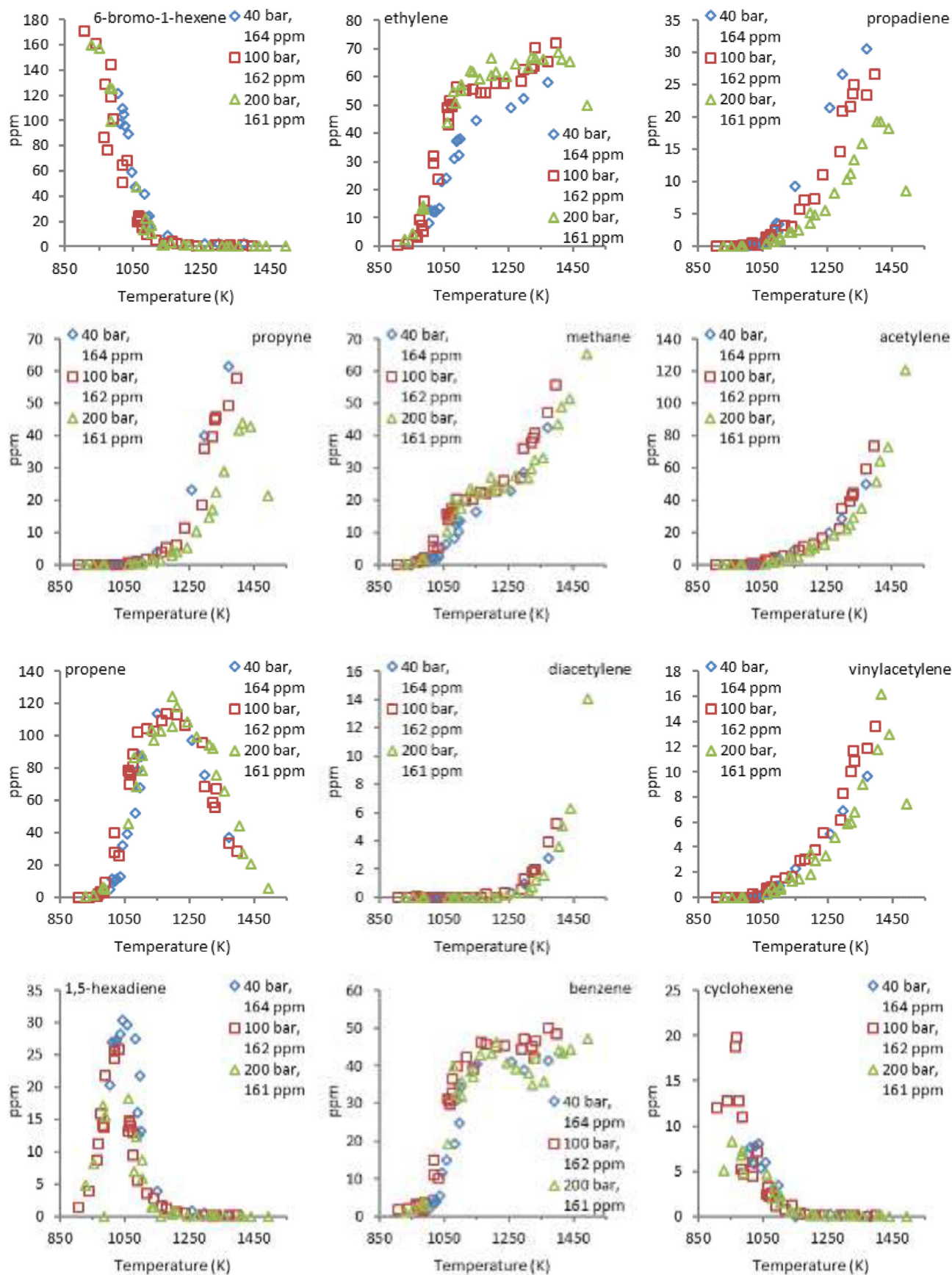


Fig. 2. Species profiles of 6-bromo-1-hexene, ethylene, propadiene, propyne, methane, acetylene, propene, diacetylene, vinylacetylene, 1,5-hexadiene, benzene, and cyclohexene obtained in 6-bromo-1-hexene pyrolysis at nominal pressures of 40, 100, and 200 bar.

benzene recovered beginning at temperatures of approximately 1250 K or higher. The 1-butene and isobutylene profiles do seem to be effected by the experimental pressure, but both these products were minor products. All the other remaining products did not exhibit any variation in their profiles due to differences in the experimental pressure.

4. Effects of initial fuel concentration on 6-bromo-1-hexene pyrolysis

The experimental results discussed so far were obtained at three different pressures but utilizing fuel mixtures that contained an approximately constant mole fraction of 6-bromo-1-hexene in the test gas. While the mole fraction of the fuel was nearly equivalent for the three sets of experiments the concentration varied depending on the experimental pressure. An additional set of experiments was conducted at 40 bar (2.27 ms mean reaction time) with 358 ppm of 6-bromo-1-hexene in the fuel mixture resulting in a fuel concentration which was approximately 218% of what was present in the previous 40 bar set which used 164 ppm of 6-bromo-1-hexene. The increase in the fuel concentration in the fuel mixture may not be exactly equivalent to the factor of 2.5 increase expected due to the increase in experimental pressure from 40 to 100 bar, but the increase of the fuel concentration in the test mixture should be sufficiently close to provide insight as to whether the product distribution differences observed were due to the difference in the initial fuel concentration present in the reaction zone or due to the variation of experimental pressure.

Figure 3 contains the species mole fraction profiles as a function of temperature of the fuel, 6-bromo-1-hexene, and the major species present in the experiments: ethylene, propadiene, propyne, methane, acetylene, propene, diacetylene, vinylacetylene, 1,5-hexadiene, benzene, and cyclohexene. All the minor species which were quantified have their profile plots included in Figure S2 in the supplemental material. Those species include: ethane, propane, toluene, cyclopropane, 1-butene, isobutylene, 1,3-cyclopentadiene, 1-hexene, and 1,3-cyclohexadiene. The compared species profiles were obtained from the 40 bar and 164 ppm of fuel, 100 bar and 162 ppm of fuel, and 40 bar and 358 ppm of fuel experiments. All the species mole fractions are normalized by the initial mole fraction of the fuel for each respective set of experiments as otherwise the amounts observed in the 40 bar and 358 ppm of fuel experiments would be significantly higher. The normalized profiles of all the product species in Fig. 3 obtained in the 40 bar and 358 ppm of fuel experiments match those in the 100 bar experiments very closely, despite the large difference in experimental pressure. This indicates that initial 6-bromo-1-hexene concentrations in the reaction zone rather than different experimental pressures influence product distributions obtained during the pyrolysis.

The major products, methane, acetylene, and propene, again have matching profiles between all three sets of experiments. In general, all the remaining species profiles match between the three sets of experiments. The most notable exceptions can be observed in benzene and isobutylene where the two 40 bar sets of experiments match more closely and have slightly lower amounts of benzene and isobutylene recovered than at 100 bar. The amount of 1-butene recovered in the 40 bar and 358 ppm of fuel experiments is lower than in the other two sets. The exact reason for this is unknown, and some of the experimental points from that set match those in the previous two sets of experimental data.

5. Comparison of 1,5-hexadiene and 6-bromo-1-hexene product distributions

The intent of using 6-bromo-1-hexene as the fuel was so that after undergoing a loss of the bromine atom the resulting

product would be the hex-5-en-1-yl radical. It is also possible for 6-bromo-1-hexene to undergo HBr loss resulting in 1,5-hexadiene. Large amounts of 1,5-hexadiene have been observed to form in all four sets of experiments presented so far, which can be seen in product distributions shown in Figs. 2 and 3. To determine the significance of the HBr loss pathway in the present work, an additional set of experiments was completed at 100 bar (2.36 ms mean reaction time) using 170 ppm of 1,5-hexadiene as the fuel allowing the product distributions obtained in the 6-bromo-1-hexene experiments at 100 bar to be directly compared with those in the 1,5-hexadiene experiments. Propadiene, propyne, and propene are expected to be among the primary products of 1,5-hexadiene pyrolysis [30] so particular attention to the comparison of those products' profiles will be given between the 1,5-hexadiene and 6-bromo-1-hexene experiments.

Figure 4 contains the species profiles of ethane, ethylene, propadiene, propyne, toluene, methane, acetylene, propene, 1-butene, vinylacetylene, 1,3-cyclopentadiene, 1,5-hexadiene, 1-hexene, benzene, and 1,3-cyclohexadiene obtained from the 100 bar pyrolysis experiments with 162 ppm of 6-bromo-1-hexene and 170 ppm of 1,5-hexadiene as the fuels. Propane, cyclopropane, diacetylene, isobutylene, and cyclohexene were not observed to form in the 1,5-hexadiene experiments at all, or formed in near zero amounts which did not produce any distinguishable profiles, so the plots for those species are omitted. The amount of propadiene and propyne recovered in the 1,5-hexadiene experiments exceeds those in the 6-bromo-1-hexene experiments, and both propadiene and propyne begin forming in greater amounts, and at lower temperatures, in the 1,5-hexadiene experiments despite 1,5-hexadiene being a more thermally stable molecule than 6-bromo-1-hexene. The ethane and 1-butene profiles obtained from the two fuel species also vary greatly, and so does the benzene profile. The abrupt formation of benzene at lower temperatures does not occur in the 1,5-hexadiene experiments. Benzene begins forming at higher temperatures and the increase in the amount recovered with respect to temperature is far more gradual. With the exception of ethylene, and propene, the remaining species profiles also vary considerably between the two fuels. The large disparity between the products observed in the 1,5-hexadiene and the 6-bromo-1-hexene experiments proves that 1,5-hexadiene was not the dominant initial product of 6-bromo-1-hexene dissociation in the present work.

6. Chemical kinetic modeling of experimental data

The Reaction Mechanism Generator (RMG) [31] was used to generate a Chemkin [32] compatible mechanism to model the hex-5-en-1-yl radical pyrolysis. RMG utilizes 45 reaction families to determine which reactions may occur for a given species based on the functional groups present in the species. The species and the functional groups are represented using adjacency lists. The mechanism generation uses a rate based algorithm originally developed by Susnow et al., [33] The mechanism generation begins by supplying RMG with a seed mechanism containing smaller species chemistry, and any other well characterized reactions which RMG may not capture correctly, and RMG progressively adds more species and reactions to the mechanism but will not overwrite those already present in the seed mechanism. RMG does not allow the generation of mechanisms involving bromine containing species making it impossible to directly generate a mechanism for 6-bromo-1-hexene. Instead, a mechanism was generated for a fuel mixture composed of 81 ppm of hex-5-en-1-yl and 80 ppm of 1,5-hexadiene with a balance of argon. 1,5-hexadiene was included in the fuel due to the substantial amounts of 1,5-hexadiene forming at the lower range of experimental temperatures covered suggesting that 6-bromo-1-hexene

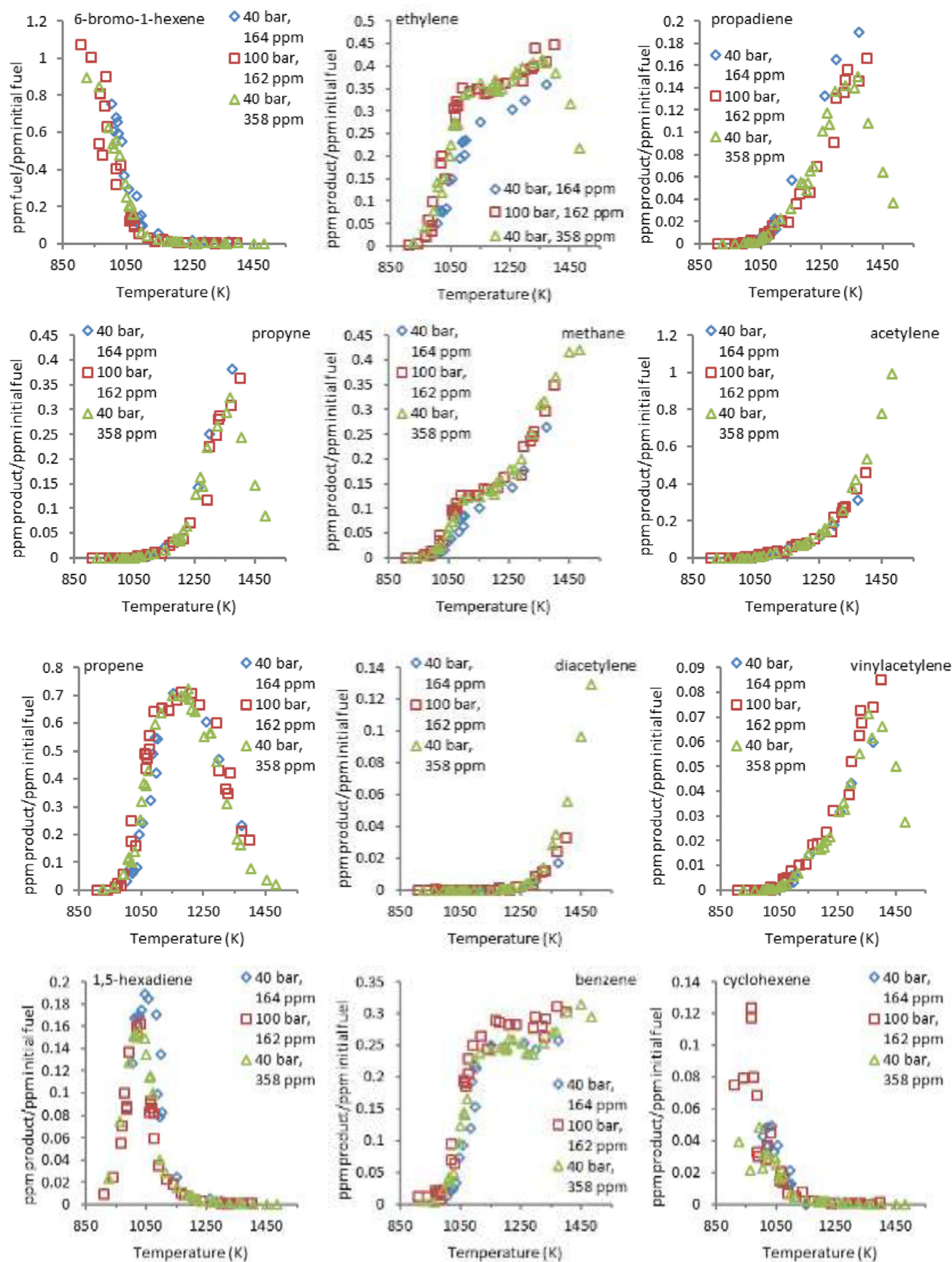


Fig. 3. Comparison of species profiles of 6-bromo-1-hexene, ethylene, propadiene, propyne, methane, acetylene, propene, diacetylene, vinylacetylene, 1,5-hexadiene, benzene, and cyclohexene obtained in 6-bromo-1-hexene pyrolysis at 40 bar and 164 ppm of 6-bromo-1-hexene, 100 bar and 162 ppm of 6-bromo-1-hexene, and 40 bar and 358 ppm of 6-bromo-1-hexene. The mole fraction of each species was normalized by the initial mole fraction of 6-bromo-1-hexene for the respective data set.

also undergoes HBr loss resulting in production of 1,5-hexadiene, and not just Br loss resulting in hex-5-en-1-yl which was desired. It is possible to manually add reactions involving 6-bromo-1-hexene to the generated mechanism, but there are no reaction rate coefficients available for 6-bromo-1-hexene pyrolysis currently in literature. The consequence of conducting the simulations directly with the radical species of interest, hex-5-en-1-yl, is that it is not representative of the physical experimental system where the fuel molecule, 6-bromo-1-hexene, has to dissociate to form the radical species over time. Peukert et al., estimated the reaction rate coefficients for the loss of iodine in 6-iodo-1-hexene pyrolysis, but he did not consider HI loss resulting 1,5-hexadiene, and one of the I-ARAS-profiles shown suggest that at temperatures > 1100 K, 6-iodo-1-hexene seems to almost completely decompose by C-I bond fission [12]. The mechanism was generated for a pressure of 200 bar with the option to generate pressure dependent reaction rate coefficients in the PLOG format as needed. The reason for generating the mechanism at the highest experimental pressure was due to the demonstrated ability to produce matching, or very closely matching, experimental data at lower pressures if the initial fuel concentration in the reaction zone is equivalent, or very close, to that in higher pressure experiments. Additionally, one of the aims of this study was to investigate whether high pressure may drive the formation of alkylcyclopentanes, so there is particular interest in modeling the high pressure chemistry.

The RMG mechanism was generated using the Chernov [34] and JetSurF2.0 [7] seed mechanisms as provided in the RMG database. Other smaller species seed mechanisms, such as the Curran pen-

tane mechanism [35] and FFCM-1 mechanism [36], were used in various combinations in previous RMG modeling efforts, but the mechanisms generated using the combination of Chernov and JetSurF2.0 seed mechanisms were found to generate the best performing mechanism in comparison to the experimental data [20,21]. The Chernov mechanism provided the needed aromatic species formation chemistry without which very little benzene formation was predicted. The JetSurF2.0 mechanism in the RMG database is not the full JetSurF2.0 mechanism, but a reduced version (by the Green Group at MIT) with majority of the reactions involving cycloalkanes being removed. All the oxygen containing species, and all the reactions involving those species were stripped from both the seed mechanisms. The removal of the oxygen containing species and their reactions was done to both reduce the size of the final RMG generated mechanism, and alleviate the computational effort needed to run Chemkin simulations. In addition to using the generated mechanism to simulate the experimental data, a Monte Carlo uncertainty analysis was performed by randomly perturbing all the “A” factors in the Arrhenius reaction rate expression by up to 30% in order to generate uncertainty bands corresponding to an uncertainty of 30% in the “A” factors in the mechanism in the same way described and used by Fridlyand [27]. Two sets of simulations were completed using the generated mechanism, one with purely hex-5-en-1-yl as the fuel and one with a fuel mixture composed of 81 ppm of hex-5-en-1-yl and 80 ppm of 1,5-hexadiene. The Monte Carlo uncertainty analysis was completed for both fuel mixtures, but is only presented for

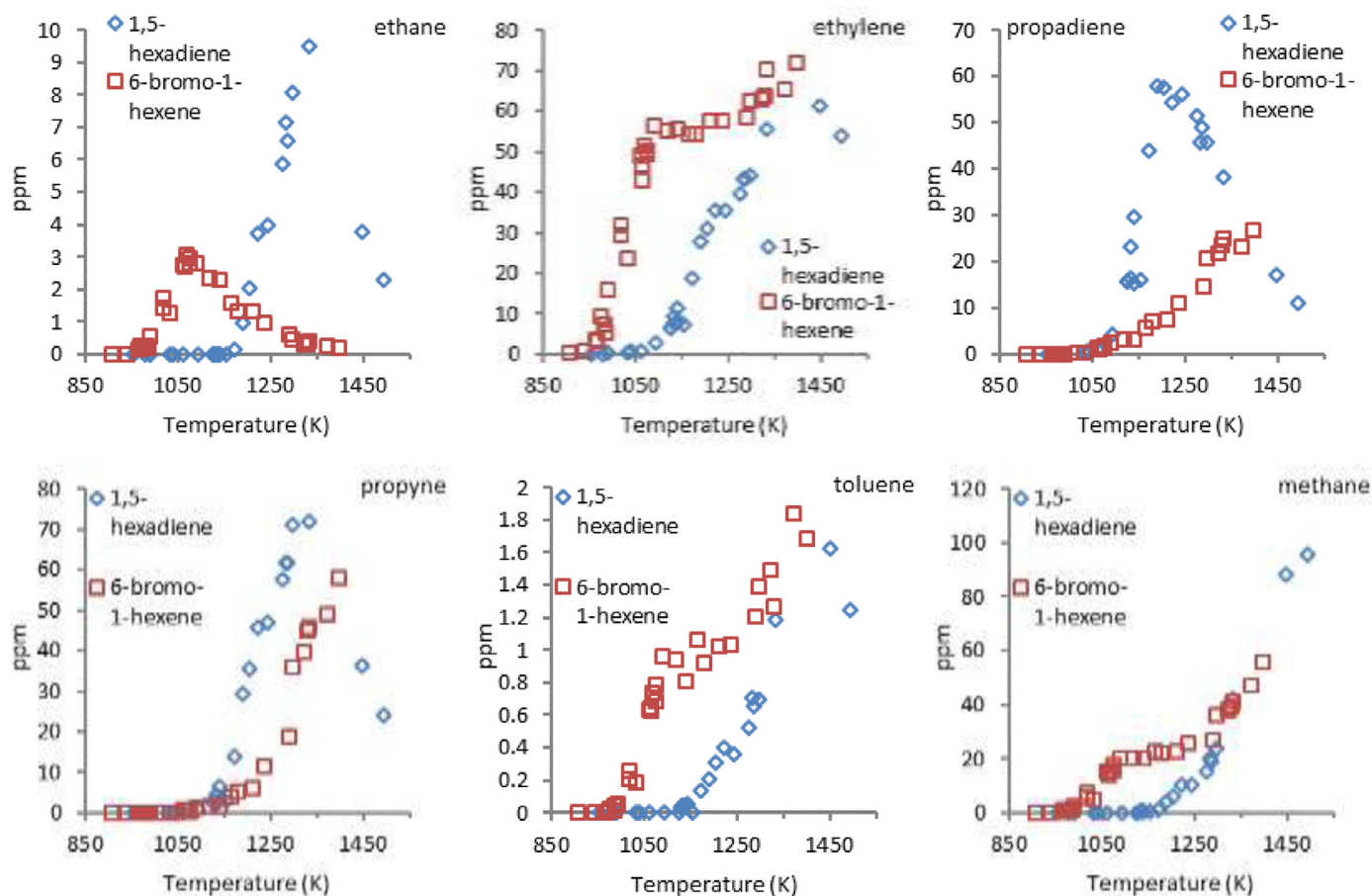


Fig. 4. Comparison of species profiles of ethane, ethylene, propadiene, propyne, toluene, methane, acetylene, propene, 1-butene, vinylacetylene, 1,3-cyclopentadiene, 1-hexene, benzene, 1,3-cyclohexadiene, and 1,5-hexadiene formed in 6-bromo-1-hexene pyrolysis at 100 bar and 162 ppm of 6-bromo-1-hexene and 1,5-hexadiene pyrolysis at 100 bar and 170 ppm of 1,5-hexadiene as the fuel.

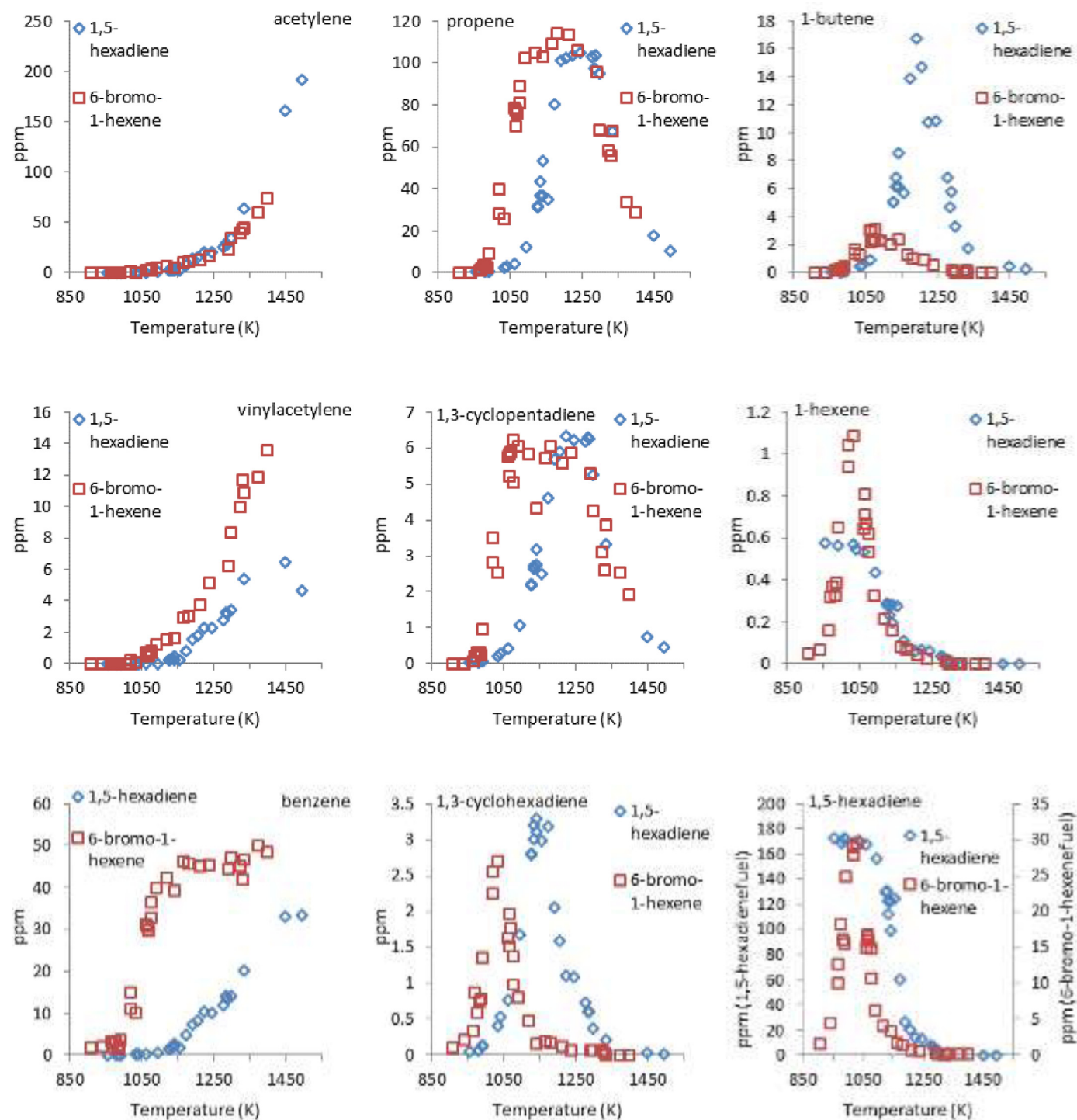


Fig. 4. Continued

the one using purely hex-5-en-1-yl as the fuel in order to retain clarity in the plots.

7. Simulation of the 6-bromo-1-hexene experimental data

Figure 5 contains the 200 bar and 161 ppm of 6-bromo-1-hexene experimental data, the simulation results for both the 161 ppm of hex-5-en-1-yl fuel and the 81 ppm of hex-5-en-1-yl with 80 ppm of 1,5-hexadiene fuel, and the uncertainty curves for the 161 ppm of hex-5-en-1-yl fuel simulation for ethane, ethylene, propene, propadiene, propyne, toluene, methane, acety-

lene, propene, 1-butene, diacetylene, isobutylene, vinylacetylene, 1,3-cyclopentadiene, 1,5-hexadiene, 1-hexene, benzene, and cyclohexene. Cyclopropane, 1,3-cyclohexadiene, and 6-bromo-1-hexene were not predicted by the mechanism so their plots are omitted. One result immediately worth noting are the simulation predictions for 1,5-hexadiene. Even for the case with only hex-5-en-1-yl as the fuel the amount of 1,5-hexadiene predicted exceeds the amount observed in the experiments. Although the peak amount of 1,5-hexadiene has not been captured in the 200 bar experiments due to lack of points near the peak, it can be estimated based

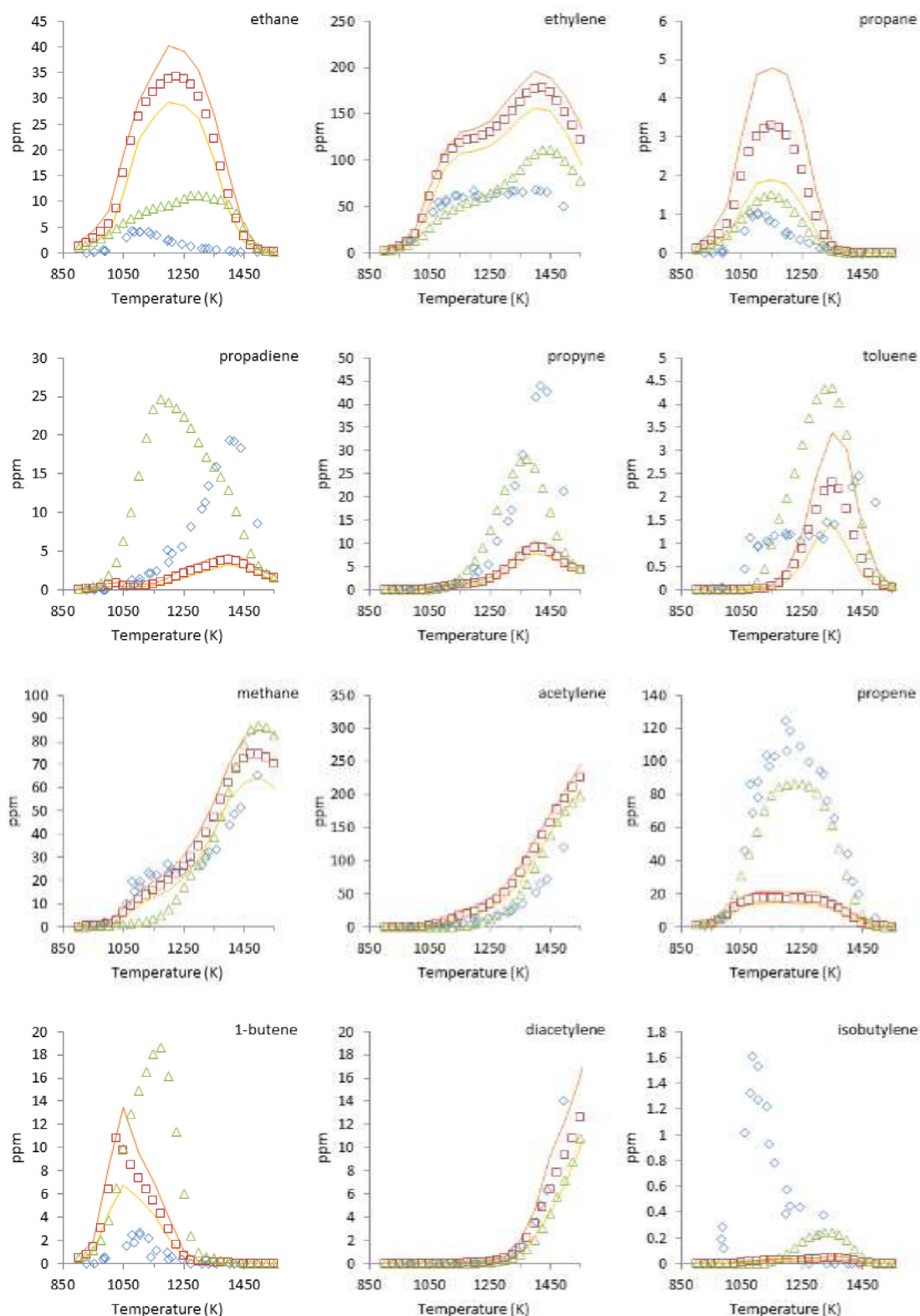


Fig. 5. Comparison of species profiles of ethane, ethylene, propane, propadiene, propyne, toluene, methane, acetylene, propene, 1-butene, diacetylene, isobutylene, vinylacetylene, 1,3-cyclopentadiene, 1,5-hexadiene, 1-hexene, benzene, and cyclohexene formed in 6-bromo-1-hexene pyrolysis at 200 bar and 161 ppm of 6-bromo-1-hexene as the fuel against the simulation results obtained from the RMG generated mechanism using 161 ppm of hex-5-en-1-yl as the fuel, and another simulation result obtained using the same mechanism but 81 ppm of hex-5-en-1-yl and 80 ppm of 1,5-hexadiene as the fuel. The uncertainty bands obtained by prescribing an uncertainty of 30% to all the A factors in the mechanism are plotted for the simulation set which was obtained with 161 ppm of hex-5-en-1-yl as the fuel. Legend: $\diamond$ experimental data, $\square$ Simulation results using 161 ppm of hex-5-en-1-yl as Fuel, $\triangle$ simulation results using 81 ppm of hex-5-en-1-yl and 80 ppm of 1,5-hexadiene as Fuel, — maximum species mole fraction from Monte Carlo analysis for the 161 ppm of hex-5-en-1-yl fuel simulation, — minimum species mole fraction from Monte Carlo analysis for the 161 ppm of hex-5-en-1-yl fuel simulation.

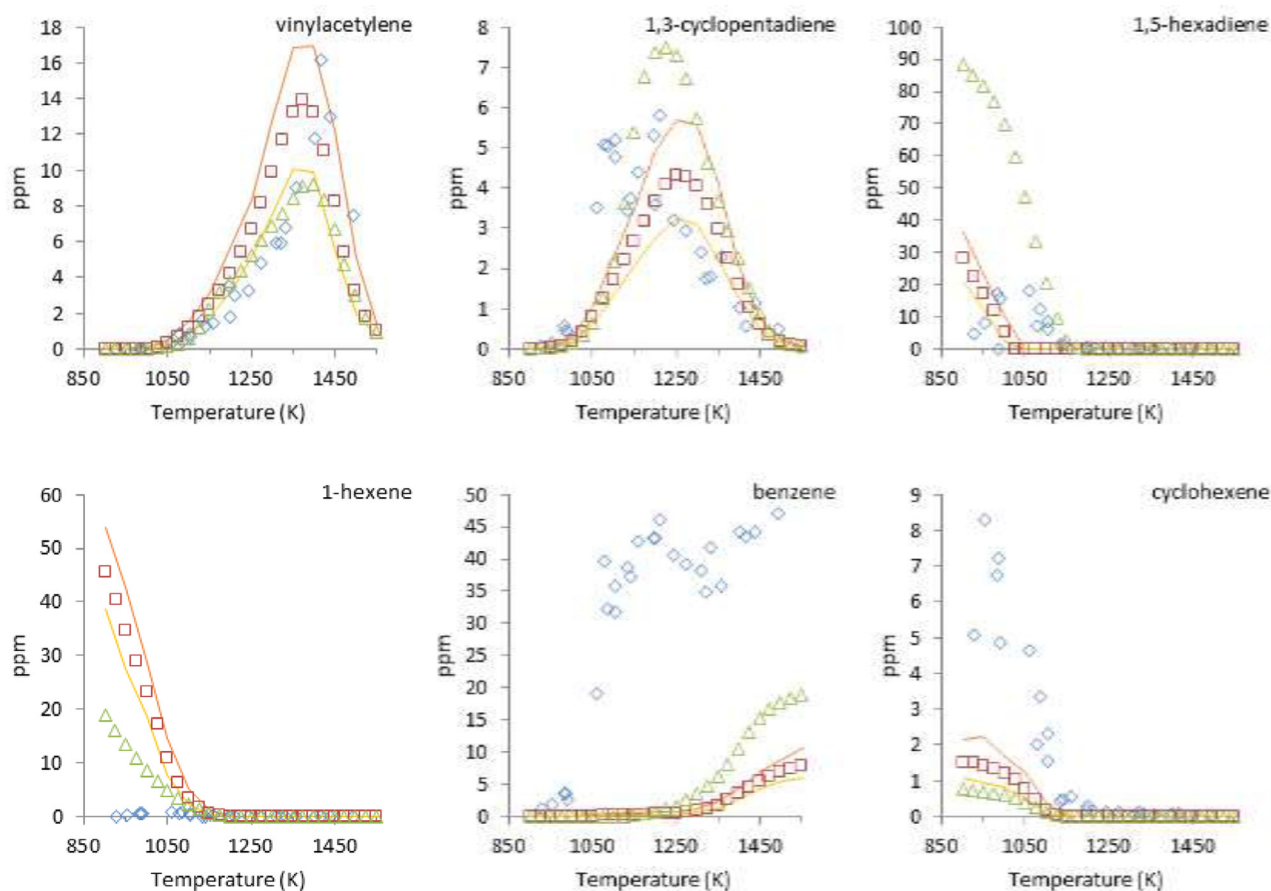


Fig. 5. Continued

on the 40 and 100 bar experiments to be around 25 ppm whereas 28 ppm is predicted if the fuel is purely hex-5-en-1-yl, and a maximum of 88 ppm is predicted for the hex-5-en-1-yl/1,5-hexadiene fuel mixture.

In general, the hex-5-en-1-yl/1,5-hexadiene fuel mixture simulation tends to provide a better prediction of the experimental data than using just the hex-5-en-1-yl radical as the fuel. This is not surprising because all of the fuel (6-bromo-1-hexene) immediately dissociating into the radical, hex-5-en-1-yl, which was used in the simulations is not a realistic scenario, and as such some of the reactions that would occur between the products at various stages of dissociation of the fuel will not be captured. The only species for which the simulation with pure hex-5-en-1-yl radical as the fuel provides a better approximation of the experimental data are for methane and vinylacetylene. The most abundant products, such as ethylene, acetylene, and particularly propene, are all better predicted by the simulation using the hex-5-en-1-yl/1,5-hexadiene fuel blend.

The large amounts of benzene observed experimentally are not captured by the simulations with either fuel. 1,3-cyclohexadiene is not predicted in the simulations at all, and cyclohexene is under predicted in both the simulations. The inability of the simulation results to capture the large amounts of benzene observed to form experimentally suggests incorrect reaction rates or missing reaction pathways in hex-5-en-1-yl pyrolysis. The Chernov seed mechanism, which is the source of the aromatic formation reactions for the RMG generated mechanism, does not contain the cyclohexene dehydrogenation pathway leading to the formation of benzene or the isomerization of fulvene to benzene. However, the inclusion of the dehydrogenation of cyclohexene pathway

would not improve the results significantly, because cyclohexene is substantially under predicted by the simulation. Recent experimental and modeling combustion studies have suggested that the hex-5-en-1-yl radical forms from the cyclohexyl radical, and then can either isomerize to the cyclopentylmethyl radical or dissociate into smaller species [1,5,6,8–10,12,37]. The isomerization of the hex-5-en-1-yl radical to the cyclohexyl radical has also been proposed [11,38]. Based on the current experimental results it seems that the isomerization of hex-5-en-1-yl to the cyclohexyl radical which then forms cyclohexene and undergoes the dehydrogenation pathway to form 1,3-cyclohexadiene and then benzene is the most probable. This hypothesis is based on the experimental observation that both cyclohexene and 1,3-cyclohexadiene decay to zero, or near zero, ppm at approximately 1150 K which is the same temperature at which benzene reaches its plateau value, until more benzene is produced at higher temperatures which is expected to be due to C₃+C₃ species recombination. This also seems to be supported by the generated model in which benzene formation sub-mechanism predominately relies on the recombination of C₃+C₃ species which is observed to occur at higher temperatures.

The cyclopentylmethyl radical could also act as an intermediate in the formation of benzene by forming either methylenecyclopentane or one of the methylcyclopentene isomers which would then dehydrogenate to fulvene and isomerize to benzene. Neither methylenecyclopentane nor and methylcyclopentene isomers were detected in this work; however, cyclopentadiene was quantified in the experiments and it could form following the cleavage of the methyl group on the methylcyclopentene isomers which would be a competing reaction pathway with the dehydrogenation to fulvene. However, cyclopentadiene formation peaks at a temperature

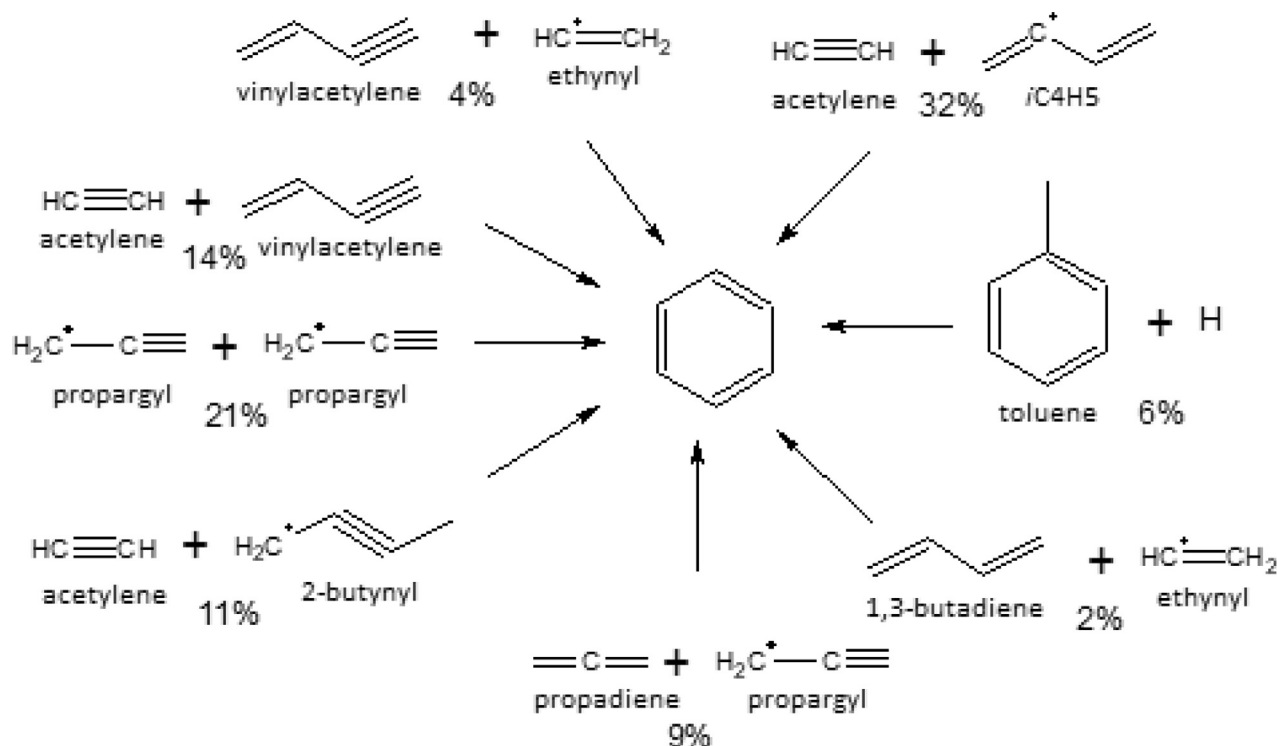


Fig. 6. Visualization of the ROP analysis of benzene formation at a temperature of 1350 K and reaction time of 2.2 ms for a simulation using 161 ppm of hex-5-en-1-yl as the fuel.

of approximately 1100 K whereas the benzene peak is not observed until about 1150 K, which would either mean that the isomerization of hex-5-en-1-yl radical to cyclohexyl radical which then forms cyclohexene that proceeds to dehydrogenate to benzene is the more favorable pathway, or that the cleavage of the methyl group in the methylcyclopentene isomers because less favorable with increasing temperature. Nonetheless, the large amount of benzene observed to form in the experiments, and the inability of the model to predict the benzene formation illustrate the need for refinement and further development of pathways in which the hex-5-en-1-yl radical acts as a benzene precursor, especially due to the current literature supporting the formation of hex-5-en-1-yl as a significant pathway in the pyrolysis of the cyclohexyl radical.

The rate of production (ROP) analysis for benzene was completed at a representative temperature of 1350 K and reaction time of 2.2 ms for the simulation using 161 ppm of hex-5-en-1-yl as the fuel, and the simulation using 80 ppm of 1,5-hexadiene and 81 ppm of hex-5-en-1-yl as the fuel. The ROP analysis was completed for benzene because it is one of the most abundant products observed experimentally which had the poorest approximation by the model, and it is also an important species in combustion chemistry. The ROP analysis will provide insight into which benzene producing reactions are included in the mechanism, and which reactions may be missing resulting in the under prediction of benzene by the model. Figure 6 illustrates the contribution of various reactions to the benzene formation for the 161 ppm of hex-5-en-1-yl fuel simulation. The C₄+C₂ species recombination reactions in total contribute 63% of the relative ROP. The propargyl radical recombination reaction has the second largest impact on benzene formation with 21% of the relative ROP. The only other C₃+C₃ recombination reaction is the recombination of propadiene and propargyl which contributes 9% of the relative ROP. The reaction of toluene with hydrogen which results in the cleavage of the methyl group contributes 6% of the relative ROP. Figure 7 visualizes the contribution of the primary benzene formation

reactions for the simulation which used 80 ppm of 1,5-hexadiene and 81 ppm of hex-5-en-1-yl as the fuel at a temperature of 1350 K and reaction time of 2.2 ms. The simulation results obtained with the fuel mixture of 80 ppm of 1,5-hexadiene and 81 ppm of hex-5-en-1-yl have resulted in a larger amount of benzene being predicted, although still substantially less than observed experimentally especially at lower temperatures, and also a larger amount of both propadiene and propyne being predicted. This is likewise reflected in the ROP analysis where the C₃+C₃ recombination reactions now have a substantially larger share, at 86% of the relative ROP. The propargyl recombination reaction alone contributes 56% of the relative ROP, and the propadiene and propargyl recombination reaction contributes 30%. The C₄+C₂ species recombination reactions now only contribute 10% of the relative ROP. The reaction of toluene with hydrogen also contributes 4% of the relative ROP.

Figure 8 illustrates the main decomposition pathways of the hex-5-en-1-yl radical as predicted by the RMG generated mechanism. The reaction pathways were obtained with the help of Chemkin Reaction Path Analyzer. The dominant decomposition pathway for both the fuel mixture simulations is the 1,4-hydrogen shift resulting in the formation of hex-5-en-4-yl. For the simulation using 161 ppm of hex-5-en-1-yl as the fuel the pathway results in nearly 91% of the relative ROP of hex-5-en-1-yl consumption and 94% for the simulation using 80 ppm of 1,5-hexadiene and 81 ppm of hex-5-en-1-yl as fuel. The second most important hex-5-en-1-yl decomposition pathway is the beta-scission of hex-5-en-1-yl which forms ethylene and the but-3-en-1-yl radical. For both fuel mixtures this pathway has approximately 4.5% of the relative ROP of hex-5-en-1-yl consumption. For the simulation using 80 ppm of 1,5-hexadiene and 81 ppm of hex-5-en-1-yl as fuel the third most important hex-5-en-1-yl radical dissociation pathway is the ring closure to form the cyclohexyl radical at approximately 1% of the relative ROP. All the remaining pathways for that fuel mixture combined have a relative ROP of less than 1%. For the

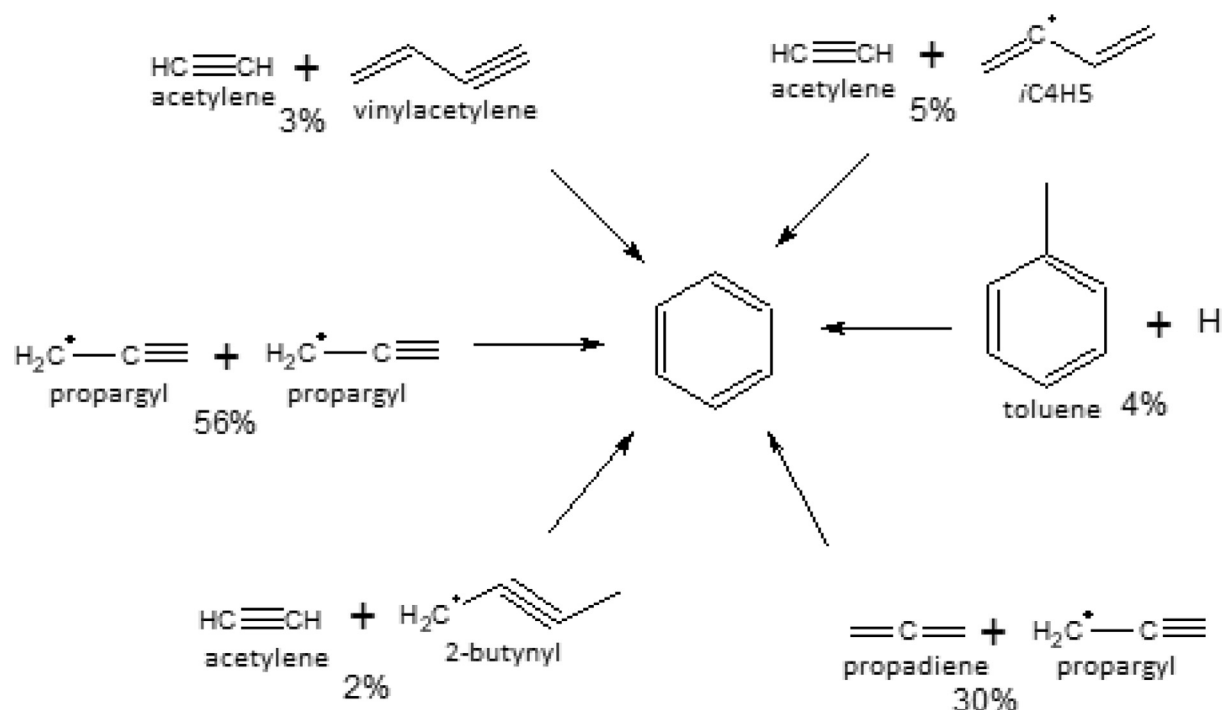


Fig. 7. Visualization of the ROP analysis of benzene formation at a temperature of 1350K and reaction time of 2.2ms for a simulation using 80 ppm of 1,5-hexadiene and 81 ppm of hex-5-en-1-yl as the fuel.

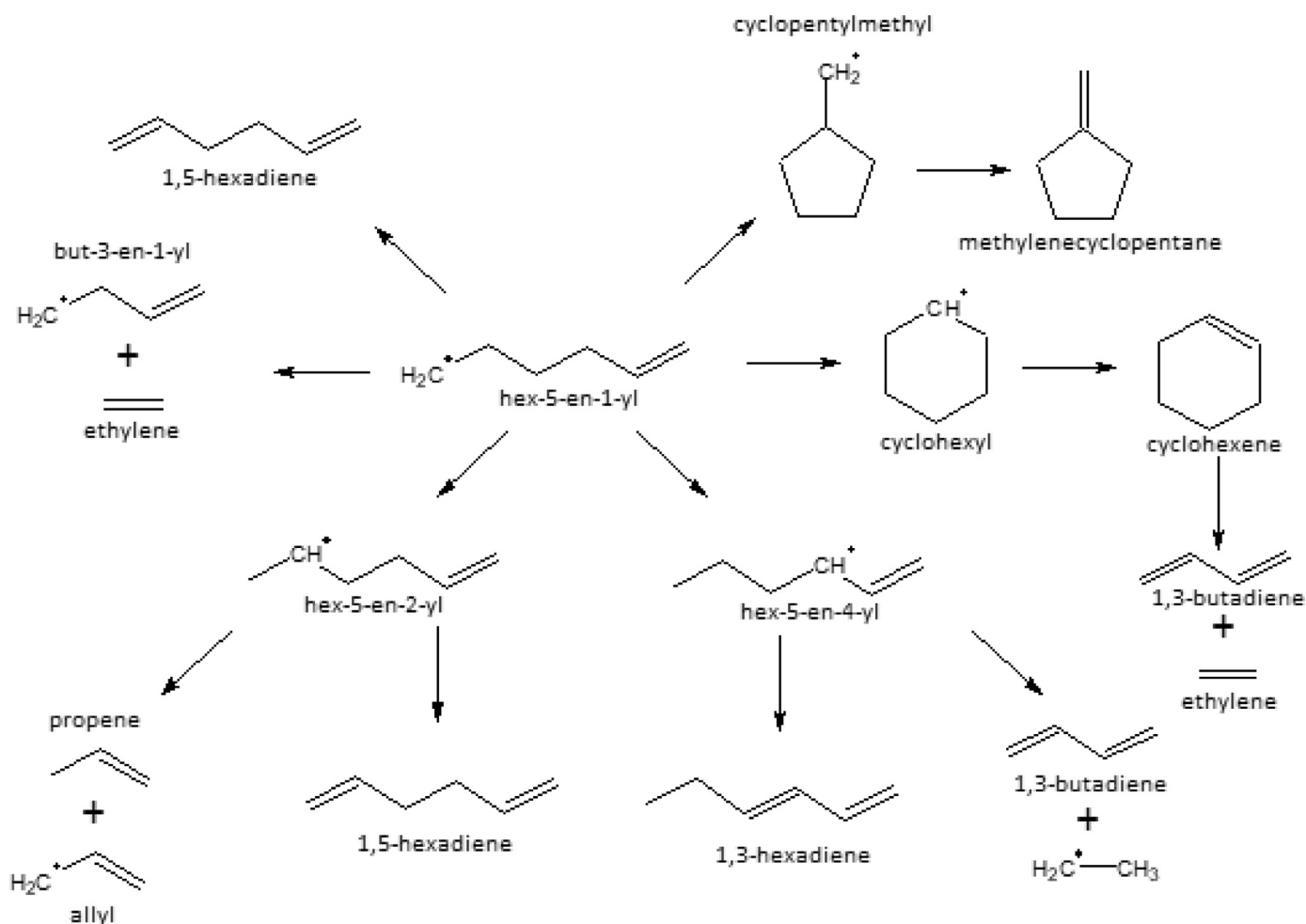


Fig. 8. Main hex-5-en-1-yl radical decomposition pathways predicted by RMG mechanism at 1350 K and 2.2 ms.

mixture utilizing 161 ppm of hex-5-en-1-yl as the fuel the third most important pathway is the formation of 1,5-hexadiene at approximately 2.5% of the relative ROP of hex-5-en-1-yl dissociation, followed by isomerization to the hex-5-en-2-yl radical or ring closure to form the cyclohexyl radical, both at approximately 1% of the relative ROP of hex-5-en-1-yl dissociation. All the remaining reactions have a contribution of less than 1% to the relative ROP of hex-5-en-1-yl dissociation. The formation of the cyclopentylmethyl radical which can then form methylenecyclopentane is also possible, but has a negligible contribution to the relative ROP of hex-5-en-1-yl radical dissociation. In both the simulation, the cyclohexyl radical will primarily form cyclohexene which will then mostly dissociate into 1,3-butadiene and ethylene. The hex-5-en-2-yl radical can either undergo beta-scission forming propene and the allyl radical, or form 1,5-hexadiene. The hex-5-en-4-yl radical can similarly undergo beta-scission forming 1,3-butadiene and the ethyl radical or form 1,3-hexadiene.

8. Conclusion

A 6-bromo-1-hexene pyrolysis study was completed at high pressures and temperatures seeking to investigate the effects of pressure on the pyrolysis of the hex-5-en-1-yl radical and also whether a high pressure environment is sufficient to drive the formation of alkylcyclopentanes which have been detected in supercritical phase pyrolysis of alkylcyclohexanes. The use of a hex-5-en-1-yl precursor allowed the study to be completed further along the reaction pathway expected to result in the formation of methylenecyclopentane, and perhaps alkylcyclopentanes, based on the proposed analogous pathway in the study of methylcyclohexane pyrolysis in the supercritical phase [16], see Fig. 1. Neither methylenecyclopentane nor any alkylcyclopentanes were observed in the current work, indicating the pressure alone is not sufficient to drive the formation of alkylcyclopentanes. Varying the experimental pressure between the three pressures of 40, 100, and 200 bar did impact the product distribution. Most noticeably the product profiles of ethane, ethylene, propane, propadiene, propyne, and toluene were affected by the variation of the experimental pressure. The experiments at all three pressures were conducted with fuel mixtures that contained approximately an equivalent mole fraction of 6-bromo-1-hexene in the test gas, but assuming ideal gas behavior, the fuel concentration in the reaction zone is expected to vary proportionally with the experimental pressure. An additional set of experiments was conducted at a pressure of 40 bar, but with a higher fuel mole fraction so that the fuel concentration in the reaction zone comes close to that in the 100 bar experiments, and the product distribution obtained matched closely that at 100 bar. This result emphasized that at the current experimental conditions the product distribution is sensitive to the initial fuel concentration.

An additional set of experiments was conducted using 1,5-hexadiene as the fuel in order to generate 1,5-hexadiene pyrolysis data at experimental conditions relevant to this study. This was done due to large amounts of 1,5-hexadiene being observed to form at low temperatures in the 6-bromo-1-hexene pyrolysis. The main intent of the 6-bromo-1-hexene pyrolysis work was to study the decomposition of the hex-5-en-1-yl radical, but if the primary reaction pathway proceeded through the HBr loss resulting in the formation of 1,5-hexadiene then the production distributions from 6-bromo-1-hexene and 1,5-hexadiene pyrolysis would be expected to be similar if not the same, but this was found not to be the case. A future study using 6-iodo-1-hexene as the fuel would be valuable in clarifying the impact of 1,5-hexadiene formation, and to further investigate the hex-5-en-1-yl radical decomposition pathways. In the present study, based on preliminary work involving iodocyclohexane and bromocyclohexane where

iodocyclohexane test gas mixtures were found to decompose by 30% overnight to cyclohexene, it was deemed that the bromine containing species are more suitable for high pressure experiments where the test gas mixtures must also be high pressure to provide adequate fill pressures for the shock tube. The 6-iodo-1-hexene studies would have to either be conducted at lower pressures, or using multiple lower pressure test gas mixtures for one set of data. The former option was not relevant to the present work where the interest was in the effect of high pressures on the product distribution, and the latter option would result in more scatter in the data due to differing fuel concentrations between mixtures and longer times necessary to conduct the experiments.

A hex-5-en-1-yl and 1,5-hexadiene chemical kinetic mechanism was generated with the help of RMG in an attempt to model the experimental data. RMG is unable to generate mechanisms for species containing bromine, hence why the mechanism was instead generated for the two main expected initial products, or hex-5-en-1-yl and 1,5-hexadiene. The mechanism was unable to accurately predict the large amounts of benzene which formed at relatively low temperatures, and it substantially under predicted cyclohexene and did not predict any 1,3-cyclohexadiene formation. The C3+C3 species recombination pathways do not seem to be adequate in capturing the benzene formation from the hex-5-en-1-yl radical. Instead, an alternative proposed pathways proceed through the isomerization of hex-5-en-1-yl radical to the cyclohexyl radical which then forms cyclohexene that undergoes dehydrogenation ultimately forming benzene. The other pathway involves the cyclopentylmethyl radical, which is the more traditionally proposed isomerization product of hex-5-en-1-yl radical. The cyclopentylmethyl radical would then either form methylenecyclopentane, or the methylcyclopentene isomers which would undergo dehydrogenation forming fulvene that would then isomerize to benzene; however, no alkylcyclopentanes, alkylcyclopentanes or alkenylcyclopentanes, which were observed in the supercritical phase pyrolysis of alkylcyclohexanes, were observed experimentally to suggest that this is pathway occurs. The large amounts of benzene observed experimentally, and the importance of hex-5-en-1-yl in the cyclohexyl radical isomerization, should act as motivation for further refinement of pathways of the hex-5-en-1-yl radical acting as a benzene precursor.

Acknowledgments

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.combustflame.2018.12.029](https://doi.org/10.1016/j.combustflame.2018.12.029).

References

- [1] C. Ji, E. Dames, B. Sirjean, H. Wang, F.N. Egolfopoulos, An experimental and modeling study of the propagation of cyclohexane and mono-alkylated cyclohexane flames, *Proc. Combust. Inst.* 33 (2011) 971–978, doi:[10.1016/j.proci.2010.06.099](https://doi.org/10.1016/j.proci.2010.06.099).
- [2] B. Sirjean, E. Dames, D.A. Sheen, F.N. Egolfopoulos, H. Wang, D.F. Davidson, R.K. Hanson, H. Pitsch, C.T. Bowman, C.K. Law, W. Tsang, N.P. Cernansky, D.L. Miller, A. Violi, R.P. Lindstedt, A High-Temperature Chemical Kinetic Model of n-Alkane, Cyclohexane, and Methyl-, Ethyl-, n-Propyl and n-Butyl-Cyclohexane Oxidation at High Temperatures, *JetSurF version 1.1*, (2009). <http://melchior.usc.edu/JetSurF/JetSurF1.1>.
- [3] P. Dagaut, M. Cathonnet, The ignition, oxidation, and combustion of kerosene: a review of experimental and kinetic modeling, *Prog. Energy Combust. Sci.* 32 (2006) 48–92, doi:[10.1016/j.pecs.2005.10.003](https://doi.org/10.1016/j.pecs.2005.10.003).
- [4] E.J. Silke, W.J. Pitz, C.K. Westbrook, M. Ribaucour, Detailed chemical kinetic modeling of cyclohexane oxidation, *J. Phys. Chem. A* 111 (2007) 3761–3775, doi:[10.1021/jp067592d](https://doi.org/10.1021/jp067592d).

- [5] Z. Hong, K.Y. Lam, D.F. Davidson, R.K. Hanson, A comparative study of the oxidation characteristics of cyclohexane, methylcyclohexane, and n-butylcyclohexane at high temperatures, *Combust. Flame* 158 (2011) 1456–1468, doi:[10.1016/j.combustflame.2010.12.019](https://doi.org/10.1016/j.combustflame.2010.12.019).
- [6] N. Liu, C. Ji, F.N. Egolopoulos, Ignition of non-premixed cyclohexane and mono-alkylated cyclohexane flames, *Proc. Combust. Inst.* 34 (2013) 873–880, doi:[10.1016/j.proci.2012.05.104](https://doi.org/10.1016/j.proci.2012.05.104).
- [7] H. Wang, E. Dames, B. Sirjean, D.A. Sheen, R. Tangko, A. Violi, J.Y.W. Lai, F.N. Egolopoulos, D.F. Davidson, R.K. Hanson, A High-Temperature Chemical Kinetic Model of n-Alkane (up to n-Dodecane), Cyclohexane, and Methyl-, Ethyl-, n-Propyl and n-Butyl-Cyclohexane Oxidation at High Temperatures, *JetSurF* version 2.0, (2010). <http://web.stanford.edu/group/haiwanglab/JetSurF/JetSurF2.0/index.html>.
- [8] C.S. McEnally, L.D. Pfefferle, Experimental study of fuel decomposition and hydrocarbon growth processes for cyclohexane and related compounds in nonpremixed flames, *Combust. Flame* 136 (2004) 155–167, doi:[10.1016/j.combustflame.2003.09.012](https://doi.org/10.1016/j.combustflame.2003.09.012).
- [9] H.R. Zhang, L.K. Huynh, N. Kungwan, Z. Yang, S. Zhang, Combustion modeling and kinetic rate calculations for a stoichiometric cyclohexane flame. 1. Major reaction pathways, *J. Phys. Chem. A* 111 (2007) 4102–4115, doi:[10.1021/jp068237q](https://doi.org/10.1021/jp068237q).
- [10] H. Li, L. Yu, S. Sun, S. Wang, X. Lu, Z. Huang, A shock tube experimental and modeling study of multicomponent gasoline surrogates diluted with exhaust gas recirculation, *Energy Fuels* 32 (2018) 3800–3813, doi:[10.1021/acs.energyfuels.7b01028](https://doi.org/10.1021/acs.energyfuels.7b01028).
- [11] D.M. Matheu, W.H. Green, J.M. Grenda, Capturing pressure-dependence in automated mechanism generation: reactions through cycloalkyl intermediates, *Int. J. Chem. Kinet.* 35 (2003) 95–119, doi:[10.1002/kin.10106](https://doi.org/10.1002/kin.10106).
- [12] S. Peukert, C. Naumann, M. Braun-Unkhoff, U. Riedel, The reaction of cyclohexane with H-atoms: a shock tube and modeling study, *Int. J. Chem. Kinet.* 44 (2012) 130–146, doi:[10.1002/kin.20595](https://doi.org/10.1002/kin.20595).
- [13] S. Granata, T. Faravelli, E. Ranzi, A wide range kinetic modeling study of the pyrolysis and combustion of naphthenes, *Combust. Flame* 132 (2003) 533–544, doi:[10.1016/S0010-2180\(02\)00465-0](https://doi.org/10.1016/S0010-2180(02)00465-0).
- [14] B. Sirjean, P.A. Glaude, M.F. Ruiz-López, R. Fournet, Theoretical kinetic study of thermal unimolecular decomposition of cyclic alkyl radicals, *J. Phys. Chem. A* 112 (2008) 11598–11610, doi:[10.1021/jp805640s](https://doi.org/10.1021/jp805640s).
- [15] W.C. Lai, C. Song, Pyrolysis of alkylcyclohexanes in or near the supercritical phase. Product distribution and reaction pathways, *Fuel Process. Technol.* 48 (1996) 1–27, doi:[10.1016/0378-3820\(96\)01030-2](https://doi.org/10.1016/0378-3820(96)01030-2).
- [16] J.F. Stewart, Supercritical Pyrolysis of the Endothermic Fuels Methylcyclohexane, Decalin and Tetralin Ph.D. thesis, Princeton University, 1999 <http://adsabs.harvard.edu/abs/1999PhDT.....388S>.
- [17] J. Kim, S.H. Park, C.H. Lee, B.-H. Chun, J.S. Han, B.H. Jeong, S.H. Kim, Coke formation during thermal decomposition of methylcyclohexane by alkyl substituted C₅ ring hydrocarbons under supercritical conditions, *Energy Fuels* 26 (2012) 5121–5134, doi:[10.1021/ef3007678](https://doi.org/10.1021/ef3007678).
- [18] J.P. Orme, H.J. Curran, J.M. Simmie, Experimental and modeling study of methyl cyclohexane pyrolysis and oxidation, *J. Phys. Chem. A* 110 (2006) 114–131, doi:[10.1021/jp0543678](https://doi.org/10.1021/jp0543678).
- [19] C.S. McEnally, L.D. Pfefferle, An experimental study in non-premixed flames of hydrocarbon growth processes that involve five-membered carbon rings, *Combust. Sci. Technol.* 131 (1998) 323–344, doi:[10.1080/00102209808935766](https://doi.org/10.1080/00102209808935766).
- [20] M.K. Liszka, K. Brezinsky, Variable high-pressure and concentration study of cyclohexane pyrolysis at high temperatures, *Int. J. Chem. Kinet.* 51 (2019) 49–73 <https://doi.org/10.1002/kin.21229>.
- [21] M.K. Liszka, K. Brezinsky, Experimental and comparative modeling study of high temperature and very high pressure methylcyclohexane pyrolysis, *Fuel* 237 (2019) 245–262 <https://doi.org/10.1016/j.fuel.2018.09.095>.
- [22] R.S. Tranter, K. Brezinsky, D. Fulle, Design of a high-pressure single pulse shock tube for chemical kinetic investigations, *Rev. Sci. Instrum.* 72 (2001) 3046–3054, doi:[10.1063/1.1379963](https://doi.org/10.1063/1.1379963).
- [23] R.S. Tranter, R. Sivaramakrishnan, N. Srinivasan, K. Brezinsky, Calibration of reaction temperatures in a very high pressure shock tube using chemical thermometers, *Int. J. Chem. Kinet.* 33 (2001) 722–731, doi:[10.1002/kin.1069](https://doi.org/10.1002/kin.1069).
- [24] Y. Hidaka, S. Shiba, H. Takuma, M. Suga, Thermal decomposition of ethane in shock waves, *Int. J. Chem. Kinet.* 17 (1985) 441–453, doi:[10.1002/kin.550170410](https://doi.org/10.1002/kin.550170410).
- [25] A. Lifshitz, K. Tamburu, F. Dubnikova, Reactions of 1-naphthyl radicals with ethylene. Single pulse shock tube experiments, quantum chemical, transition state theory, and multiwell calculations, *J. Phys. Chem. A* 112 (2008) 925–933, doi:[10.1021/jp077289s](https://doi.org/10.1021/jp077289s).
- [26] A. Matsugi, K. Yasunaga, H. Shiina, Thermal decomposition of 1,1,1-trifluoroethane revisited, *J. Phys. Chem. A* 118 (2014) 11688–11695, doi:[10.1021/jp510227k](https://doi.org/10.1021/jp510227k).
- [27] A. Fridlyand, S.S. Goldsborough, K. Brezinsky, S.S. Merchant, W.H. Green, Influence of the double bond position on the oxidation of decene isomers at high pressures and temperatures, *Proc. Combust. Inst.* 35 (2015) 333–340, doi:[10.1016/J.PROCI.2014.06.020](https://doi.org/10.1016/J.PROCI.2014.06.020).
- [28] A. Fridlyand, S.S. Goldsborough, K. Brezinsky, Chemical kinetic influences of alkyl chain structure on the high pressure and temperature oxidation of a representative unsaturated biodiesel: methyl nonenoate, *J. Phys. Chem. A* 119 (2015) 7559–7577, doi:[10.1021/acs.jpca.5b00914](https://doi.org/10.1021/acs.jpca.5b00914).
- [29] A. Comandini, T. Malewicki, K. Brezinsky, Online and offline experimental techniques for polycyclic aromatic hydrocarbons recovery and measurement, *Rev. Sci. Instrum.* 83 (2012) 34101, doi:[10.1063/1.3692748](https://doi.org/10.1063/1.3692748).
- [30] A. Fridlyand, P.T. Lynch, R.S. Tranter, K. Brezinsky, Single pulse shock tube study of allyl radical recombination, *J. Phys. Chem. A* 117 (2013) 4762–4776, doi:[10.1021/jp402391n](https://doi.org/10.1021/jp402391n).
- [31] C.W. Gao, J.W. Allen, W.H. Green, R.H. West, Reaction mechanism generator: automatic construction of chemical kinetic mechanisms, *Comput. Phys. Commun.* 203 (2016) 212–225, doi:[10.1016/j.cpc.2016.02.013](https://doi.org/10.1016/j.cpc.2016.02.013).
- [32] Reaction Design, ANSYS CHEMKIN-Pro 18.0, (2016). http://www.reactiondesign.com/support/help/help_usage_and_support/how-to-cite-products/.
- [33] R.G. Susnow, A.M. Dean, W.H. Green, P. Peczak, L.J. Broadbelt, Rate-based construction of kinetic models for complex systems, *J. Phys. Chem. A* 101 (1997) 3731–3740, doi:[10.1021/jp9637690](https://doi.org/10.1021/jp9637690).
- [34] V. Chernov, M.J. Thomson, S.B. Dworkin, N.A. Slavinskaya, U. Riedel, Soot formation with C1 and C2 fuels using an improved chemical mechanism for PAH growth, *Combust. Flame* (2014), doi:[10.1016/j.combustflame.2013.09.017](https://doi.org/10.1016/j.combustflame.2013.09.017).
- [35] J. Bugler, B. Marks, O. Mathieu, R. Archuleta, A. Camou, C. Grégoire, K.A. Heufer, E.L. Petersen, H.J. Curran, An ignition delay time and chemical kinetic modeling study of the pentane isomers, *Combust. Flame* 163 (2016) 138–156, doi:[10.1016/j.combustflame.2015.09.014](https://doi.org/10.1016/j.combustflame.2015.09.014).
- [36] G.P. Smith, Y. Tao, H. Wang, Foundational Fuel Chemistry Model Version1.0 (FFCM-1), (2016). <http://nanoenergy.stanford.edu/ffcm1>.
- [37] B. Sirjean, F. Buda, H. Hakka, P.A. Glaude, R. Fournet, V. Warth, F. Battin-Leclerc, M. Ruiz-Lopez, The autoignition of cyclopentane and cyclohexane in a shock tube, *Proc. Combust. Inst.* 31 (2007) 277–284, doi:[10.1016/j.proci.2006.07.247](https://doi.org/10.1016/j.proci.2006.07.247).
- [38] T. Gierczak, J. Gawłowski, J. Niedzielski, Isomerization of excited cyclohexyl, 1-hexen-6-yl and cyclopentylmethyl radicals, *J. Photochem. Photobiol. A Chem.* 55 (1990) 25–35, doi:[10.1016/1010-6030\(90\)80015-P](https://doi.org/10.1016/1010-6030(90)80015-P).